

Create a Microsoft Lakehouse

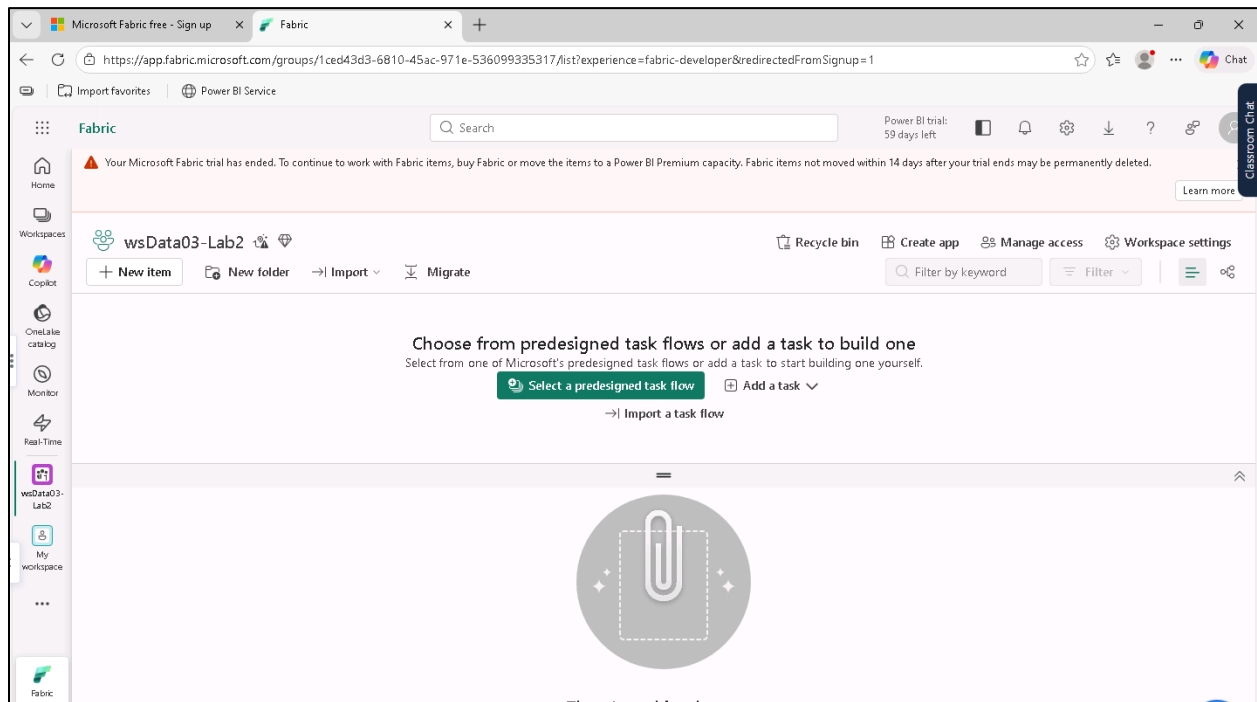
Certification Track: DP-600 [Fabric Analytics Engineer Associate]

This lab shows you how to create a Microsoft Fabric Lakehouse to help you understand the foundation for data engineering and analytics.

1. Create Workspace

This is the starting point for any data project in Fabric. The first step is to create a workspace to hold and organize all your Fabric items.

- Open Microsoft Fabric in your web browser and go to the Fabric home page:
<https://app.fabric.microsoft.com/home?experience=fabric>
- Select Workspace from the left menu.
- Click +New workspace.
- Name your workspace and select Create. Your new workspace should be empty.

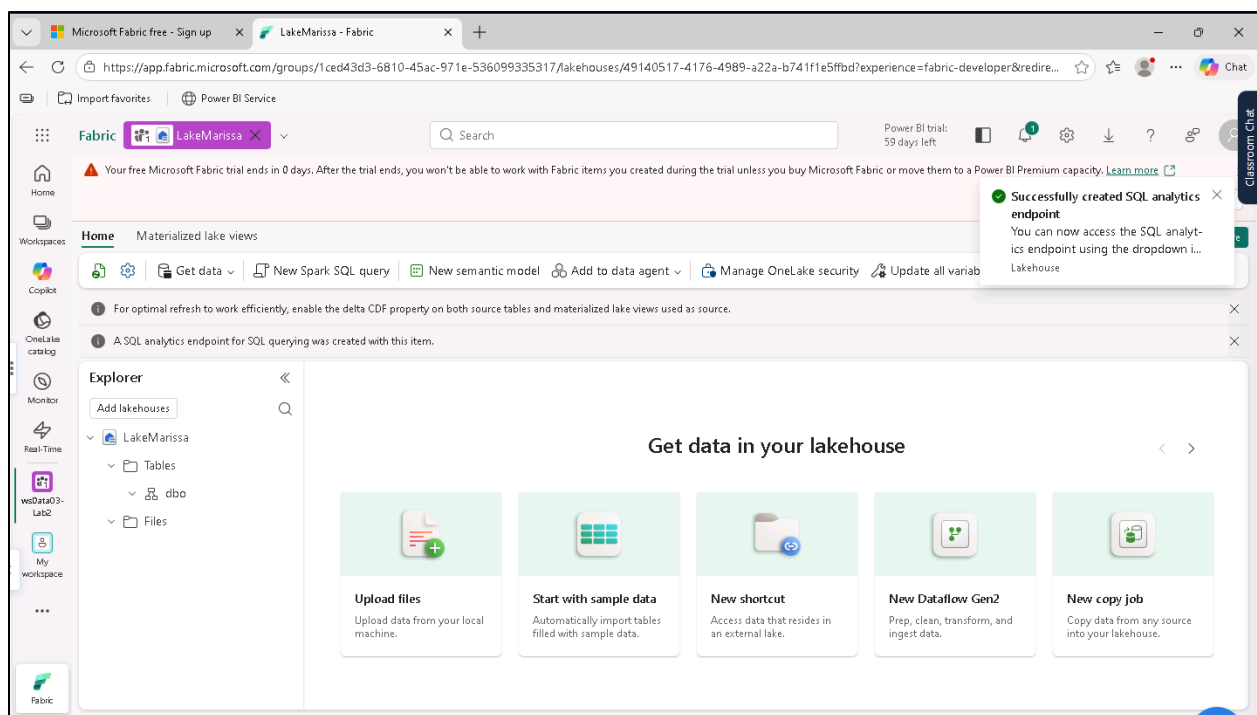


2. Create Lakehouse

A Lakehouse is an architecture which brings a data lake and a data warehouse together that provides flexibility and performance at the same time. It allows storing of raw data and runs SQL on structured tables.

The second step is to add a Lakehouse to your workspace to store data. You will see that a new Lakehouse is created with empty Files and Tables folders.

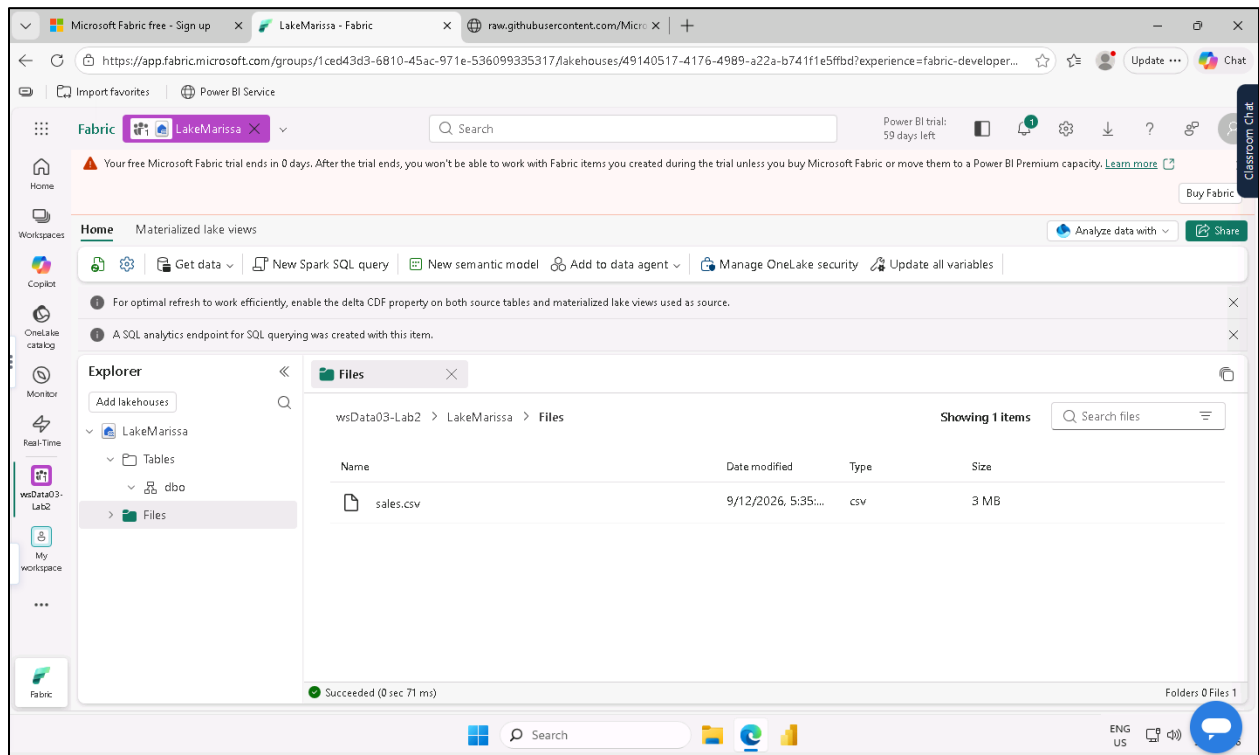
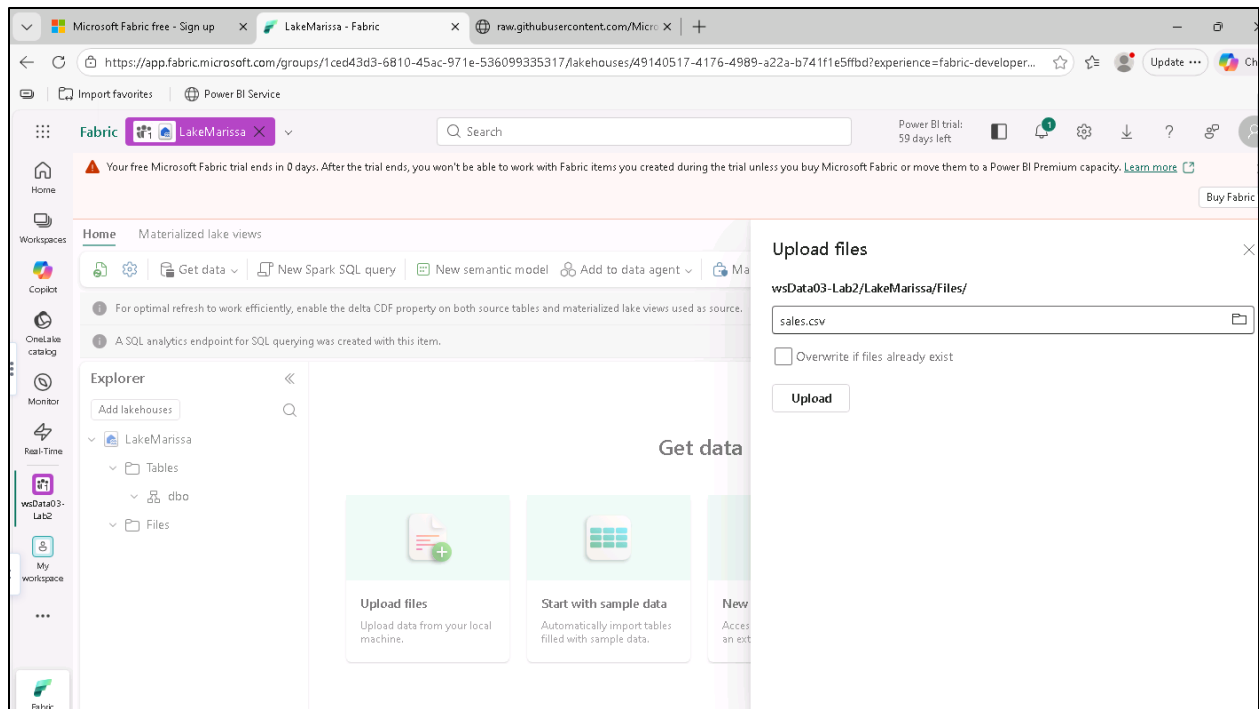
- Inside your workspace select +New item.
- Choose Lakehouse.
- Enter a name for the Lakehouse.
- Click Create.



3. Upload a File with Sample Data

Once your Lakehouse is set up, the next step is to load data into it so it can be accessed for analytics.

- Download the CSV file and save it on your computer as sales.csv
- Create a folder in your Lakehouse. Go to the Explorer pane-> Files folder, open the ... menu -> New subfolder. Name the folder data.
- Upload the CSV. Open the ... menu on the data folder-> Upload-> Upload files. Select sales.csv from your computer.
- Verify and preview. Open Files/data & confirm sales.csv is there. Select the file to preview its contents.



4. Explore Shortcuts

The purpose of a shortcut is to keep one copy of data. A shortcut in Microsoft Fabric is a pointer to data stored somewhere else either internal or an external storage. Because the data isn't copied, shortcuts prevent data duplication and ensures you always access data from its original source to get the most current, up-to-date information.

- Click the ... menu for the Files folder and select New shortcut.
- View the available data source types for shortcuts. We are just exploring so close the New shortcut dialog box without creating a shortcut.

5. Load File Data into a Table

You can work with data files directly using Apache Spark. But to query the data with SQL, you need to load the data from the file into a table. Fabric will automatically create a Delta table which adds support for ACID transactions similar to a data warehouse.

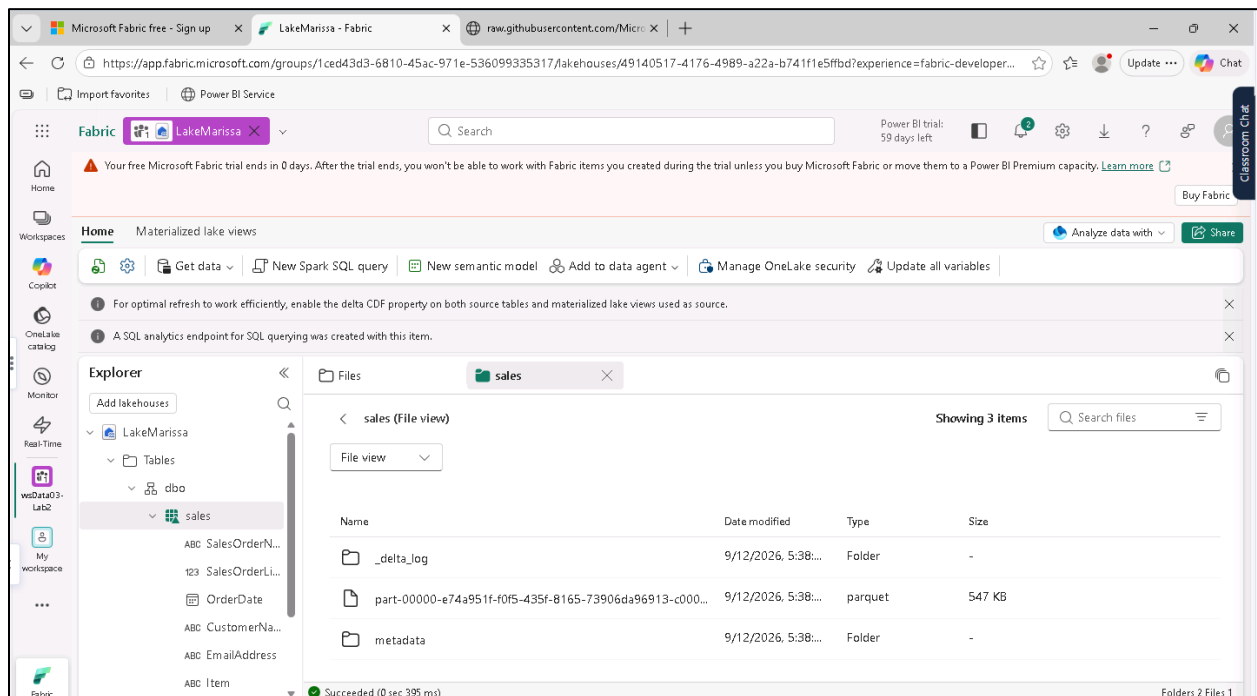
- From Explorer pane, open the uploaded file ex. sales.csv
- In the ... menu for the file, choose Load to Tables-> New table.
- In Load to table dialog box, select the target table name.
- Confirm the load operation and wait for the table to be created and loaded.
- In the Explorer pane, check the Tables section to see the new Delta table called sales.
- In the ... menu for the sales table, select File View to see the underlying files for this table.

The screenshot displays the Microsoft Fabric Lakehouse interface. The Explorer pane on the left shows the hierarchy: LakeMarissa > Tables > dbo > sales. The main area shows the 'sales' table in 'Table view' with 1,000 rows displayed. The table has columns: SalesOrderNumber, OrderDate, CustomerName, EmailAddress, Item, Quantity, UnitPrice, and Total. The status bar at the bottom indicates 'Succeeded (20 sec 708 ms)' and 'Columns: 9 Rows: 1,000'.

	abc SalesOrderN...	123 SalesOrderLi...	OrderDate	abc CustomerN...	abc EmailAddress	abc Item	123 Quantity	12 UnitPrice	12 Ta
1	SO43701	1	2019-07-01	Christy Zhu	christy12@adve...	Mountain-100 Si...	1	3399.99	271.99
2	SO43704	1	2019-07-01	Julio Ruiz	julio1@adventur...	Mountain-100 Bl...	1	3374.99	269.99
3	SO43705	1	2019-07-01	Curtis Lu	curtis9@adventu...	Mountain-100 Si...	1	3399.99	271.99
4	SO43700	1	2019-07-01	Ruben Prasad	ruben10@adven...	Road-650 Black, ...	1	699.0982	55.927
5	SO43703	1	2019-07-01	Albert Alvarez	albert7@advent...	Road-150 Red, 62	1	3578.27	286.26
6	SO43697	1	2019-07-01	Cole Watson	cole1@adventur...	Road-150 Red, 62	1	3578.27	286.26
7	SO43699	1	2019-07-01	Sydney Wright	sydney61@adve...	Mountain-100 Si...	1	3399.99	271.99

The process of loading a file into a table results in a delta table being created. A Delta table in Fabric is just a folder in OneLake that contains two kinds of files:

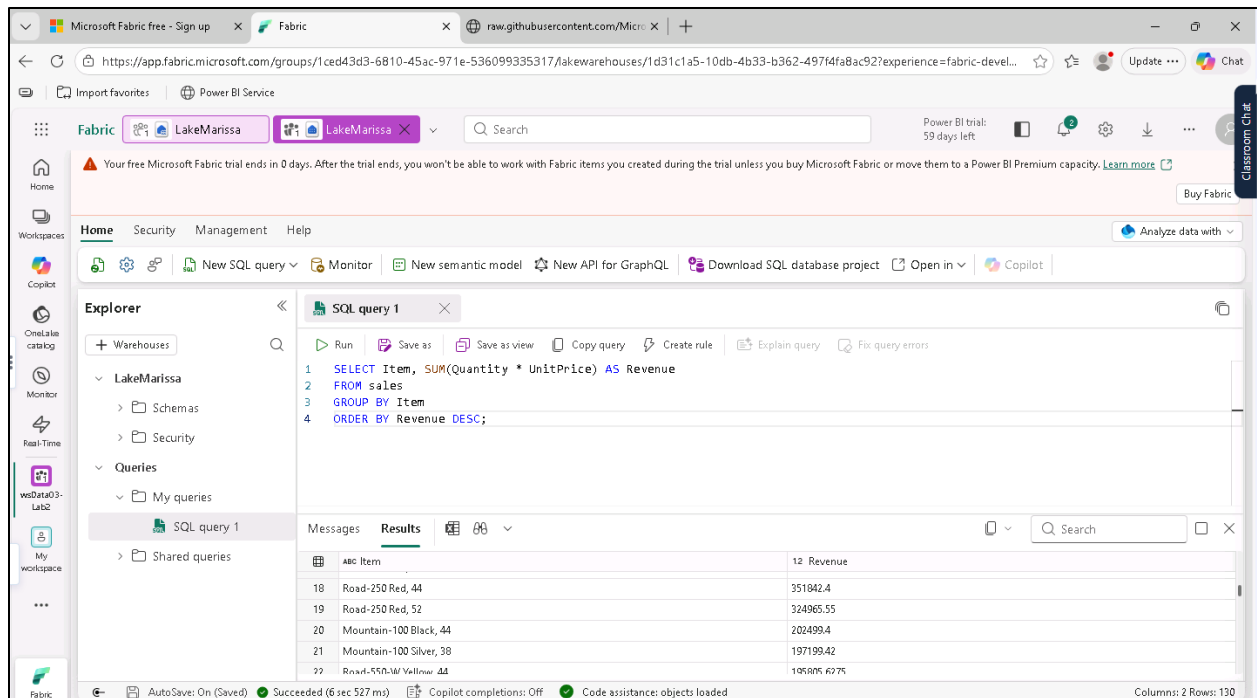
- 1) Parquet files that store the raw data. Parquet files are column-based files designed for efficiency.
- 2) A transaction log (_delta_log folder) that tracks every change to the table. This allows for data versioning and time travel.



6. Use SQL to Query Tables

When you create a Lakehouse and add tables to it, an SQL endpoint is automatically created where you can use SQL to query the tables. This is a fast, serverless, read-only interface to run standard SQL SELECT statements without using Spark.

- Click the New SQL query button to open a new query editor.
- Type in the SQL query.
- Click the Run button to run query and view the results.



7. Create a Visual Query

While SQL is common for many data professionals, Power BI users can apply their Power Query skills to create visual queries using drag and drop and no code is needed.

- Open a new visual query
- Add the sales table
- Choose the columns you want and click OK
- Group the data
- Review the results

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https://app.fabric.microsoft.com/groups/1ced43d3-6810-45ac-971e-536099335317/akewarehouses/1d31c1a5-10db-4b33-b362-4974fa8ac92?experience=fabric-devel... ☆ ☆ Update ... Chat

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Explorer Warehouses LakeMarissa Schemas dbo Tables sales Views Functions Stored Proce INFORMATION_ queryinsights

SQL query 1 SQL query 2 Visual query 1

Manage columns Reduce rows Sort Transform Combine Save as view View SQL

Source Database Table

Download Excel file Visualize results

	SalesOrderNumber	SalesOrderLine	OrderDate	CustomerName	EmailAddress	Item	Quantity	UnitPr
1	SO43701		7/1/2019	Christy Zhu	christy12@adventure-works.com	Mountain-100 Silver, 44	1	
2	SO43704		7/1/2019	Julio Ruiz	julio1@adventure-works.com	Mountain-100 Black, 48	1	
3	SO43705		7/1/2019	Curtis Lu	curtis9@adventure-works.com	Mountain-100 Silver, 38	1	
4	SO43700		7/1/2019	Ruben Prasad	ruben10@adventure-works.com	Road-650 Black, 62	1	
5								

Completed (3.67 s) Columns: 9 Rows: 99+

Copy SQL connection string

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https://app.fabric.microsoft.com/groups/1ced43d3-6810-45ac-971e-536099335317/akewarehouses/1d31c1a5-10db-4b33-b362-4974fa8ac92?experience=fabric-devel... ☆ ☆ Update ... Chat

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SQL query 1 SQL query 2 Visual query 1

Manage columns Reduce rows Sort Transform Combine Save as view View SQL

Source Database Table

Download Excel file Visualize results

Choose columns

Choose the columns to keep.

Search

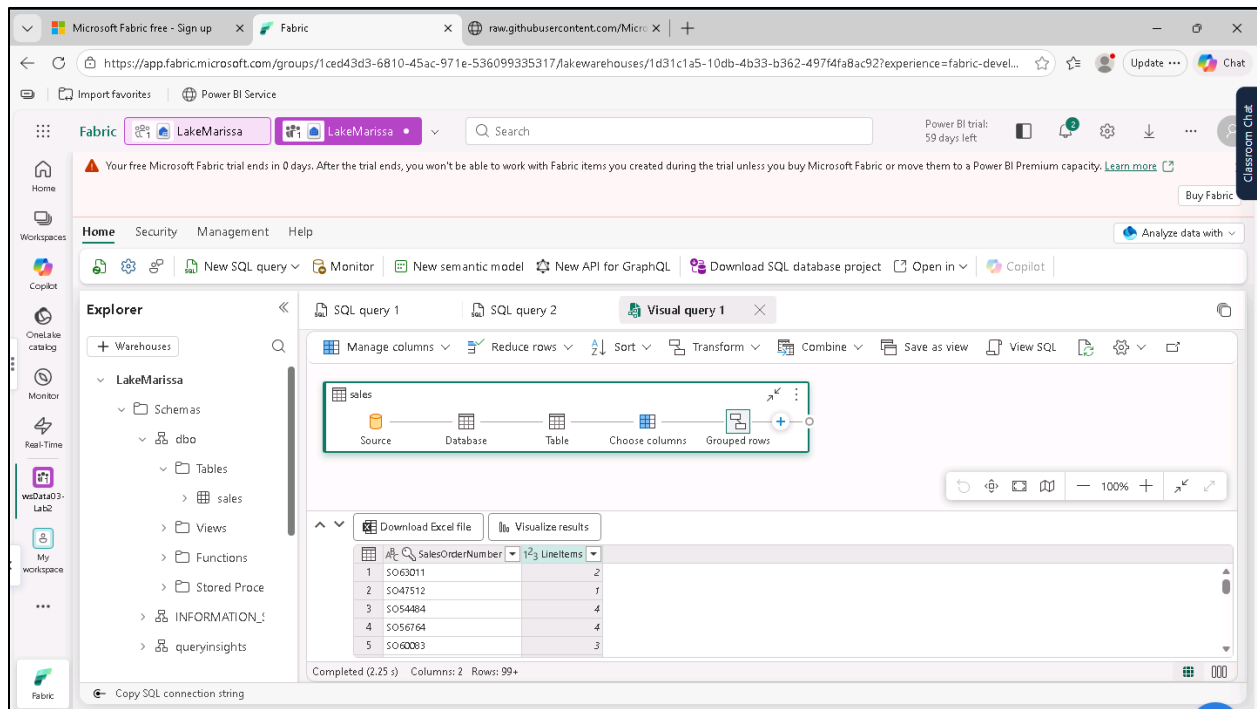
- ☒ (Select all)
- ☒ SalesOrderNumber
- ☒ SalesOrderLine
- ☐ OrderDate
- ☐ CustomerName

OK Cancel

	SalesOrderNumber	SalesOrderLine	OrderDate	CustomerName	EmailAddress	Item	Quantity	UnitPr
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Columns: 9 Rows: 99+

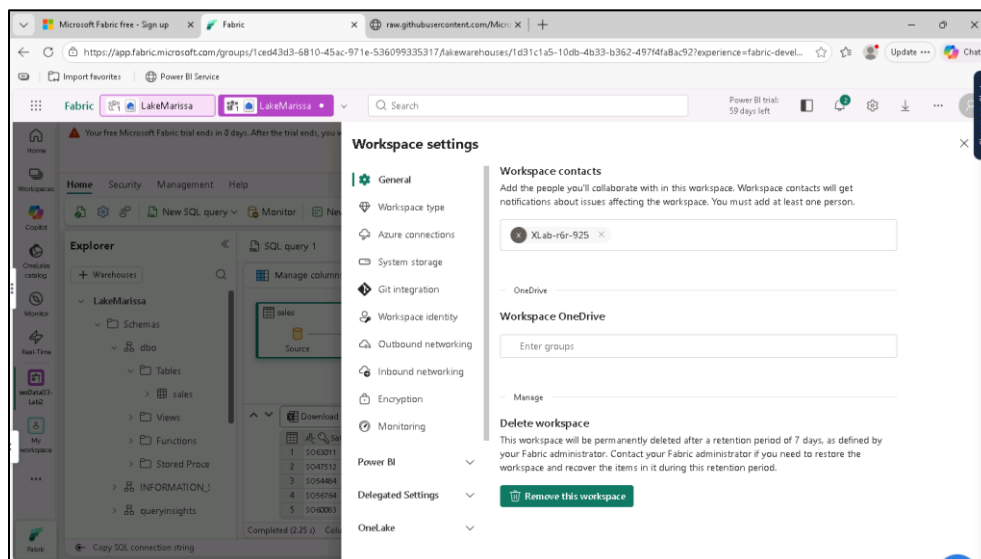
Copy SQL connection string



8. Clean up Resources

It is important to delete the resources after completing the lab to stop unnecessary capacity consumption which incur costs.

- Select the icon for your workspace on the left to view all items in the workspace.
- In the toolbar, select Workspace settings.
- In the General section, select Remove this workspace.



9. Reflection

In this lab, I learned that a Lakehouse is the core storage used in the Fabric ecosystem. You start by creating a workspace, then build a Lakehouse inside the workspace, and finally upload a data file into the Lakehouse and create tables. I also learned that SQL can only be used once raw data in files are converted into structured tables and that the SQL endpoint provides a simple way to query the tables without needing Spark, this is the unique function of Fabric. In addition, it is easy to create visual queries which requires no coding. The lab made it clear that Lakehouse's are the essential foundation for data engineering and analytics.